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Problem Solving Protocol

multi-type neighbors enhanced global topology and pairwise attribute learning for drug–protein interaction prediction

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Abstract

Motivation: Accurate identification of proteins interacted with drugs helps reduce the time and cost of drug development. Most of previous methods focused on integrating multisource data about drugs and proteins for predicting drug–target interactions (DTIs). There are both similarity connection and interaction connection between two drugs, and these connections reflect their relationships from different perspectives. Similarly, two proteins have various connections from multiple perspectives. However, most of previous methods failed to deeply integrate these connections. In addition, multiple drug-protein heterogeneous networks can be constructed based on multiple kinds of connections. The diverse topological structures of these networks are still not exploited completely.
Results: We propose a novel model to extract and integrate multi-type neighbor topology information, diverse similarities and interactions related to drugs and proteins. Firstly, multiple drug–protein heterogeneous networks are constructed according to multiple kinds of connections among drugs and those among proteins. The multi-type neighbor node sequences of a drug node (or a protein node) are formed by random walks on each network and they reflect the hidden neighbor topological structure of the node. Secondly, a module based on graph neural network (GNN) is proposed to learn the multi-type neighbor topologies of each node. We propose attention mechanisms at neighbor node level and at neighbor type level to learn more informative neighbor nodes and neighbor types. A network-level attention is also designed to enhance the context dependency among multiple neighbor topologies of a pair of drug and protein nodes. Finally, the attribute embedding of the drug-protein pair is formulated by a proposed embedding strategy, and the embedding covers the similarities and interactions about the pair. A module based on three-dimensional convolutional neural networks (CNN) is constructed to deeply integrate pairwise attributes. Extensive experiments have been performed and the results indicate GCDTI outperforms several state-of-the-art prediction methods. The recall rate estimation over the top-ranked candidates and case studies on 5 drugs further demonstrate GCDTI's ability in discovering potential drug-protein interactions.

Keywords: multiple drug–protein heterogenous networks, drug–protein interaction, three-dimensional convolutional neural network, attentions at neighbor-type level and at network level, multi-type neighbors enhanced global topology, graph neural network with neighbor node-level attention

Introduction

The discovery of drug–protein interactions (DTI) is crucial for drug development as drugs interact with the corresponding proteins to exert their actions [1, 2]. Accurate identification of potential drug–protein interactions reduces the cost and time required for drug development [3, 4]. The computational prediction method of DTIs have demonstrated the ability to screen reliable candidate proteins for further experimental verification [5–7].

Early DTIs prediction methods are divided into ligand-based methods and docking-based methods. The former utilizes the similarities between protein ligands to predict drug–protein interactions [8, 9]. The latter is based on the three-dimensional structure information of a protein and dynamic simulation to estimate the probability of drug–protein interaction [10]. Due to the ligands and three-dimensional structures of some proteins are unavailable, the application of the methods is difficult [11].

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Machine learning techniques [12] are utilized for model learning. Luo *et al.* [13] identified potential interactions between drugs and proteins from an established heterogeneous network using singular value decomposition (SVD) algorithm. Several other methods obtained the topological relationships of drugs and proteins based on random walk on the drug-protein network [13–15]. However, methods based on matrix factorization and random walk cannot extract the deep features of drugs and proteins. Bleakley *et al.* and Keum *et al.* proposed a method that integrates the support vector machine (SVM) model with the bipartite local model to predict the protein interacted with a given drug [16, 17]. A prediction model based on gradient boosting decision tree (GBDT) was also established for predicting new DTIs [18, 19]. These methods only use one kind of drug similarity and one kind of protein similarity, and thus they did not exploit the multiple kinds of drug and protein similarities and other data related drugs and proteins from multiple data sources to reveal the complex relationships among them.

Recently, deep learning-based methods also have been proposed and applied on DTI prediction [20], circRNA-disease association prediction [21] and DNA-binding proteins identification [22]. Deep learning technology has achieved good performance in learning complex networks [23, 24], so it has also been applied to integrate drug-related and protein-related data from multiple sources to obtain more reliable prediction results. Some methods have been developed to infer protein candidates for each drug based on convolutional neural network [5, 25, 26]. A method based on path-category feature extraction and random forest is proposed for DTIs prediction [27]. Based on non-negative matrix factorization (NMF) and dilation convolution, Xuan *et al.* proposed a DTIs prediction method [28]. Zheng *et al.* proposed a model that combines CNN with long and short-term memory (LSTM) network [29]. Ezzat *et al.* proposed a novel DTIs prediction model based on matrix factorization, and they utilized singular value decomposition algorithm to decompose the DTI matrix into protein and drug feature matrices [30]. However, these methods fail to apply to DTIs prediction with multi-type neighbor nodes. A prediction model based on graph convolutional network (GCN) and deep neural network was constructed [31]. Wang *et al.* [32] utilized the principal component analysis (PCA) to extract drug- and protein-related features and predicted DTIs by the long short-term memory (LSTM) network. A method learns the node representations by deep walks for DTI prediction [33]. Lv *et al.* proposed a link prediction method based on drug-protein interaction local structural similarity (DLS) [34]. These methods ignore the drug and protein similarities which are important auxiliary information for prediction. A prediction model based on graph convolutional autoencoder and generative adversarial network is established [35]. Manoochehri *et al.* established a model based on semibipartite graph [36] for predicting the DTIs. The prediction model based on convolutional neural networks and network representation learning is constructed [37]. These methods only use one kind of drug similarity and one kind of protein similarity. There are multiple similarities between drugs, including chemical structure similarity and functional similarity based on DTIs, etc [19]. Similarly, there are multiple similarities between proteins. Diverse drug (protein) similarities can reflect the intrarelations between drug (protein) nodes from different perspectives. However, most of the previous methods ignore the diverse similarities of drugs and proteins.

In this article, a new method for predicting candidate drug-protein interactions, GCDTI, is proposed. The contributions of the model include the following:

- Based on DTIs, drug similarities, drug interactions, protein similarities, and protein interactions, four drug-protein heterogeneous networks are constructed separately. Each network is composed of inter- and intrarelations between nodes to embed the similarities and interactions across multiple data sources, including drugs and proteins. Multi-Type neighbor node sequences of drugs and proteins are formed by applying random walk on each heterogeneous network to reflect the neighbor topological information.
- Since all the neighbors of each drug and protein node in the heterogeneous network contribute differently to DTIs prediction, we propose a neighbor node level attention mechanism to obtain informative neighbors of each node, which are then encoded by graph neural network.
- The multi-type neighbor topological features of a drug (protein) node are adaptively fused by an attention mechanism at neighbor type level. Since the neighbor topological representations of drug-protein pairs obtained separately from multiple networks are interdependent, we further define a network level attention mechanism to learn the global topological representation.
- We establish a pairwise embedding mechanism to integrate multiple similarities and interactions of a drug-protein pair to form the three-dimensional attribute embedding. A framework based on multi-layer three-dimensional CNN is proposed to obtain and learn attribute representations of node pairs. The improved prediction results are compared with seven DTIs prediction models and demonstrated on case studies of five drugs.

Materials and Methods

In order to predict the interaction of a protein with a given drug, a drug–protein interaction prediction model consisting of two branches (Figure 1) is proposed. In the first branch, for the similarities and interactions of

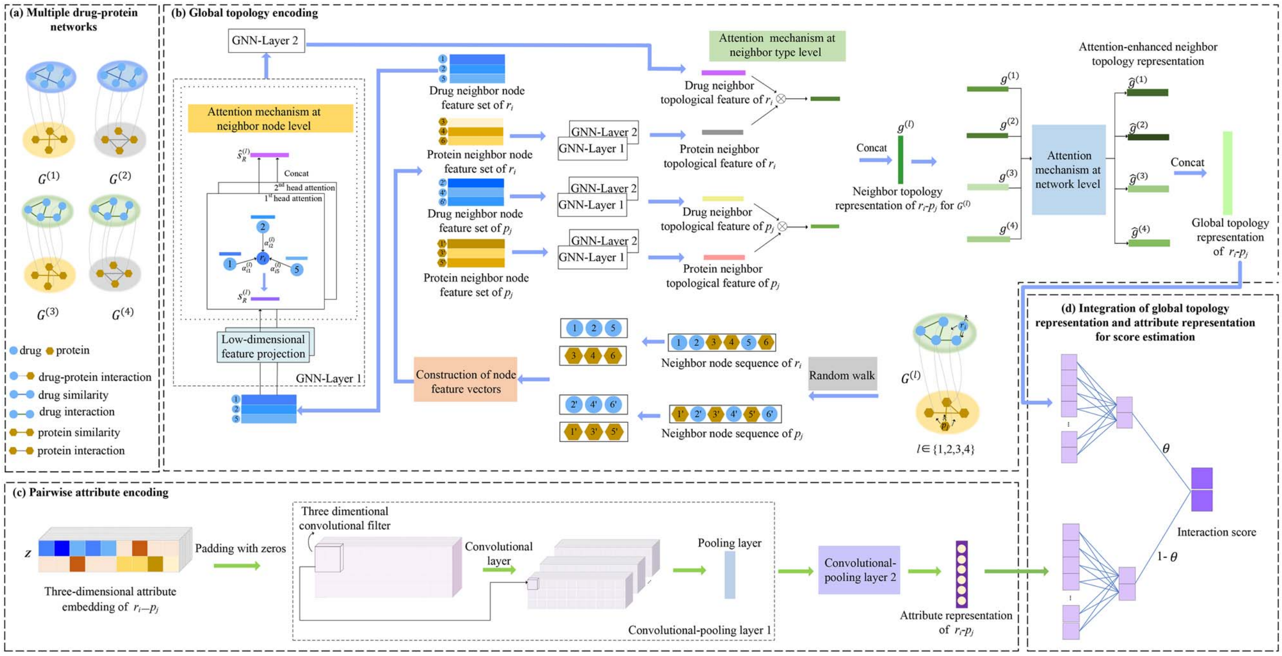

Figure 1. Framework of the proposed GCDTI model. **(A)** construct four drug-protein networks based on drug-related and protein-related data. **(B)** learn the global topology representation by random walk, GNN with attention, and two additional attention mechanisms. **(C)** encode pairwise attribute based on three-dimensional convolutional neural networks. **(D)** integrate global topology representation and attribute representation for score estimation.

drugs and proteins, we construct multiple heterogeneous networks composed of drug and protein nodes. The multi-type neighbor information of drugs and proteins is encoded by random walk and graph neural network with neighbor node level attention mechanism. We design a novel neighbor-type level and a network-level attention mechanism to encode and learn the global topological representation of a pair of drug and protein nodes. The second branch constructs a framework based on three-dimensional convolutional neural network to encode the attribute representation of a drug-protein pair. Finally, the drug–protein interaction score is weighted by the interaction scores of the two branches, which reflects the possibility of a drug interacting with a protein.

Dataset

Drug-protein interactions, drug similarities of chemical structure, drug-drug interactions, protein similarities of sequence, and protein-protein interactions are obtained from a published paper [13] for DTIs prediction. We originally extract 1923 known DTIs from DrugBank database [38], covering 708 drugs and 1512 proteins. The 10 036 drug–drug interactions and 7363 protein–protein interactions are originally obtained from DrugBank database and Human Protein Reference Database (HPRD) [39], respectively. Based on chemical structure and sequence, the similarities of drugs and proteins are calculated, respectively.

Construction of multiple drug–protein heterogeneous networks

Matrix representations of multiple connections among drugs and proteins

For drug–protein interactions, drug similarities of chemical structure, drug–drug interactions, protein similarities of sequence and protein–protein interactions, we construct four drug-protein heterogeneous networks $G = \{G^{(1)}, G^{(2)}, G^{(3)}, G^{(4)}\}$ as shown in Figure 2. The l-th drug-protein heterogeneous network $G^{(l)}$, $l \in \{1, 2, 3, 4\}$ is composed of a node set $V = \{V_R \cup V_P\}$. N_r drug nodes and N_p protein nodes form sets $V_R = \{r_1, r_2, ..., r_{N_r}\}$ and $V_P = \{p_1, p_2, ..., p_{N_p}\}$, respectively. According to the types of edge-connected nodes in the network, the connections between nodes are divided into the inter-relation between drugs and proteins of different types of nodes, and the intrarelations between nodes of the same type, such as drug–drug and protein–protein.

The drug–protein inter-relation matrix is defined as A^{R-P} which represents the interactions between N_r drugs and N_p proteins. If drug $r_i \in V_R$ interact with protein $p_j \in V_P$, $(A^{R-P})_{ij} = 1$. Otherwise, $(A^{R-P})_{ij} = 0$.

There are two intrarelations between drugs, including chemical structure similarities and interactions, and the two intrarelation matrices for drugs are denoted by R as follows:

$$R = \begin{cases} R^{che} = (R^{che})_{ij} \in \mathbb{R}^{N_r \times N_r} \\ \quad\quad if\ r_i \in V_R, r_j \in V_R \\ R^{int} = (R^{int})_{ij} \in \mathbb{R}^{N_r \times N_r} \\ \quad\quad if\ r_i \in V_R, r_j \in V_R \end{cases} \quad (1)$$

where R^{che} records the chemical structure similarities between N_r drugs. Based on the biological premise that similar drugs have more common chemical structures and the known chemical structure vectors of drugs, we use Tanimoto coefficient [40] as provided by [13] to calculate R^{che}. The interactions between N_r drugs are represented by R^{int}. $(R^{int})_{ij} = 1$ means r_i and r_j interact with each other. Otherwise, no interaction is observed.

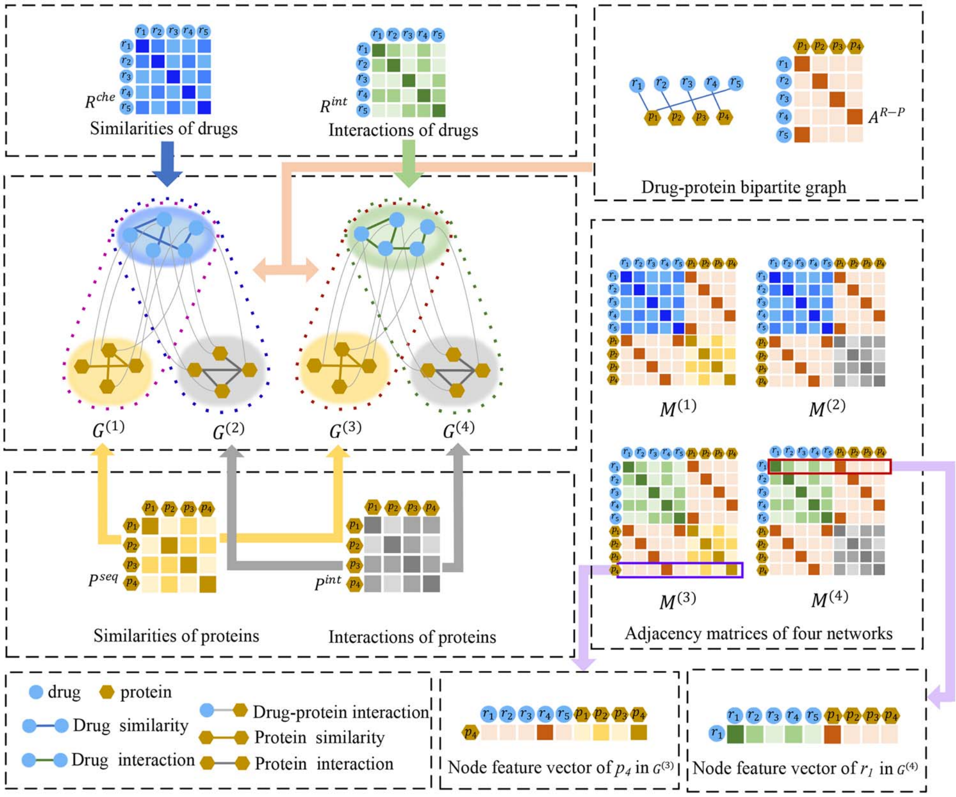

Figure 2. Construction of multiple drug–protein heterogeneous networks based on multiple kinds of connections.

Similarly, there are two intrarelations between proteins: similarities and interactions. The two intrarelation matrices for proteins are defined as:

$$P = \begin{cases} P^{seq} = (P^{seq})_{ij} \in \mathbb{R}^{N_p \times N_p} \\ \qquad \text{if } p_i \in V_P, p_j \in V_P \\ P^{int} = (P^{int})_{ij} \in \mathbb{R}^{N_p \times N_p} \\ \qquad \text{if } p_i \in V_P, p_j \in V_P \end{cases} \tag{2}$$

where P^{seq} contains the sequence similarities between N_p proteins. If the sequences of two proteins are more similar, their functions are usually more similar. Therefore, the similarity between the i-th and the j-th protein is calculated by Smith-Waterman score [41] as provided by [13]. We use P^{int} to represent the interactions between N_p proteins. When p_i and p_j are observed to have an interaction, $(P^{int})_{ij} = 1$. Otherwise, $(P^{int})_{ij} = 0$.

Based on the inter-relation matrix A^{R-P} between drugs and proteins, intrarelation matrices R of drugs and intrarelation matrices P of proteins, we construct the adjacency matrix $M^{(l)} \in \mathbb{R}^{(N_r+N_p)\times(N_r+N_p)}$ of the l-th drug–protein network $G^{(l)}$,

$$M^{(l)} = \begin{bmatrix} R^{t_1} & A^{R-P} \\ (A^{R-P})^T & P^{t_2} \end{bmatrix} \tag{3}$$

where $t_1 \in \{che, int\}$, $t_2 \in \{seq, int\}$. When $l = 1$, t_1 is *che* and t_2 is *seq*. When $l = 2$, t_1 is *che* and t_2 is *int*. When $l = 3$, t_1 is *int* and t_2 is *seq*, and when $l = 4$, t_1 is *int* and t_2 is *int*. $(A^{R-P})^T$ is the transpose matrix of A^{R-P}.

Node feature vectors of drugs and proteins

The node features of drugs and proteins are important auxiliary information for predicting the proteins interacted with drugs. Since the process of representing the node feature vector of a drug or a protein in each network is similar, r_i's node feature vector in the first network is taken as an example to describe this process. The matrices R^{che}, P^{seq} and A^{R-P} form the adjacency matrix $M^{(1)}$ of the first network. R^{che} represents the chemical structure similarities between N_r drugs, the i-th row $(R^{che})_i$ represents the node feature vector of r_i at the similarity level of chemical structure. A^{R-P} records the interactions between N_r drugs and N_p proteins, and the node feature vector of r_i at drug-protein interaction level is represented by the i-th row of A^{R-P} which is defined as $(A^{R-P})_i$. $(R^{che})_i$ and $(A^{R-P})_i$ record the node feature of r_i from different levels. Therefore, $(R^{che})_i$ and $(A^{R-P})_i$ are connected as the node feature vector of r_i, represented by $x_i^{(1)} \in \mathbb{R}^{(N_r+N_p)}$. Similarly, the node feature vector $h_j^{(1)} \in \mathbb{R}^{(N_r+N_p)}$ of p_j is formed by connecting the j-th row of $(A^{R-P})^T$ and P^{seq}.

Therefore, we can extract the node feature vector $x_i^{(l)} \in \mathbb{R}^{(N_r+N_p)}$ and $h_j^{(l)} \in \mathbb{R}^{(N_r+N_p)}$ of r_i and p_j in the l-th drug–protein network separately.

Global topology encoding by integrating multi-type neighbor topology from multiple networks

Neighbor node sequence extraction based on random walk

For the established drug–protein network $G = \{G^{(1)}, G^{(2)}, G^{(3)}, G^{(4)}\}$, each network reflects the topological relations related to drug and protein nodes from different perspectives. Therefore, to make full use of the topological structures in the four drug-protein networks, a drug (protein) node's sequence composed of drug and protein neighbor nodes is formulated through the node2vec strategy [42] to assist in topological encoding.

The random walker can start walking from any drug or protein node in a drug–protein network. For example, the walker starts at drug r_i to node u in the l-th drug–protein network $G^{(l)}$. We take u as the neighbor node of r_i and as a new starting point. Then, the normalized probability of a walker from node u to its next neighbor node v is divided into the following three cases: When there is an edge between r_i and v, the walker turns to the node v with a probability of $w_{uv} \times \frac{1}{K}$; w_{uv} is the value of the connection (interaction or similarity) between node u and v. K is the normalizing constant. When there is no edge between r_i and v, the probability of a walker turning from u to v is $w_{uv} \times \frac{1}{m} \times \frac{1}{K}$; m is a hyperparameter used to control the probability of the walker jumping from u to its one-hop neighbor. A smaller m means that the walker is more likely to jump from u to its one-hop neighbor v. When r_i and v are the same node, the walker walks to v with a probability of $w_{uv} \times \frac{1}{n} \times \frac{1}{K}$; n is also a hyperparameter that controls the probability of immediately revisiting a node in the walk. The smaller the parameter n, the more likely the walker returns to the original starting node r_i.

After repeating N_l times of the above-mentioned random walk process, a sequence of drug and protein neighbor nodes with length N_l of r_i is formed. Similarly, we can also obtain N_t number of neighbor node sequences of r_i. We count the occurrence frequency of each neighbor node of r_i in these sequences and sort them in descending order. A drug node has multi-type neighbor nodes: drug neighbor nodes and protein neighbor nodes. Neighbor nodes of different types reflect the topological features of the node from different perspectives. Therefore, a node sequence composed of only drug or protein neighbor nodes is separately formulated. For $G^{(l)}$, r_i's drug (protein) neighbor node sequence $Y_R^{(l)}(Y_P^{(l)})$ consists of the top N_s drug (protein) neighbor nodes.

Learning multi-type neighbor topological features based on GNN with attention

Due to the drug-type neighbors and protein-type neighbors of each drug and protein node express different topological information, we establish a graph neural network framework with an attention mechanism at neighbor node level to learn the multi-type neighbor topological features of a drug and a protein.

The process of learning drug neighbor topological feature and protein neighbor topological feature of a drug or a protein is similar. Therefore, we describe this process by learning drug neighbor topological feature of r_i in the l-th drug-protein network $G^{(l)}$. Each drug neighbor of r_i is $r_j \in Y_R^{(l)}$ and $x_j^{(l)}$ is the node feature vector of r_j. For all drug neighbors of r_i, the corresponding node feature vectors form the drug neighbor node feature set of r_i. To obtain a denser node feature, $x_j^{(l)}$ is projected into a low-dimensional feature space to form a low-dimensional node feature vector $\widehat{x}_j^{(l)}$,

$$\widehat{x}_j^{(l)} = W_{nei} x_j^{(l)} \tag{4}$$

where $W_{nei} \in \mathbb{R}^{N_f \times (Nr+Np)}$ is the weight matrix, and N_f denotes the dimension of a low-dimensional feature space. Similarly, the low-dimensional node feature vector $\widehat{x}_i$ of r_i is obtained.

As all drug neighbors of r_i contribute differently to the learning of the topological feature of r_i, we propose an attention mechanism at neighbor node level to identify informative neighbors. The attention score at neighbor node level is $e_{ij}^{(l)}$,

$$e_{ij}^{(l)} = LeakyReLU(a_{nei}^T [\, \widehat{x}_i^{(l)} \,||\, \widehat{x}_j^{(l)}]) \tag{5}$$

where *LeakyReLU* is a nonlinear activation function [43], $e_{ij}^{(l)}$ denotes the importance of drug neighbor r_j for learning the topological feature of r_i, and || denotes the connection operation. a_{nei}^T is used to capture the context relationships between multiple drug neighbors of r_i. The normalized attention weight $\alpha_{ij}^{(l)}$ for the drug neighbor r_j of r_i is represented by:

$$\alpha_{ij}^{(l)} = \frac{exp(e_{ij}^{(l)})}{\sum_{o \in Y_R^{(l)}} exp(e_{io}^{(l)})} \tag{6}$$

where *exp* is an exponential function. Finally, the topological feature vector $s_R^{(l)}$ of r_i's drug neighbors is obtained by aggregating the node features of all the drug neighbors with attention weights as follows:

$$s_R^{(l)} = \sigma(\sum\nolimits_{j \in Y_R^{(l)}} \alpha_{ij}^{(l)} \widehat{x}_j^{(l)}) \tag{7}$$

where σ represents a sigmoid nonlinear activation function. In order to reduce the bias in the learning process of the attention mechanism at neighbor node level, we extend it to multiple heads. We repeat the aboved learning process N_{head} times and connect the learned topological features from beginning to end to form the final topological feature vector $\widehat{s}_R^{(l)}$ of r_i's drug neighbors,

$$\widehat{s}_R^{(l)} =_{head=1}^{N_{head}} ||(s_R^{(l)})_{head} \tag{8}$$

Similarly, the final topological feature vector $\widehat{s}_P^{(l)}$ of r_i's protein neighbors is obtained in the l-th network $G^{(l)}$.

Integration of the topological features of multi-type neighbors based on neighbor-type level attention

The drug neighbor topological feature vector $\widehat{s}_R^{(l)}$ and the protein neighbor topological feature vector $\widehat{s}_P^{(l)}$ of r_i, which are learned from the l-th network $G^{(l)}$, contribute differently to the topological representation of r_i.

Therefore, a neighbor type level attention mechanism is established to obtain the enhanced neighbor topological representation of r_i. The attention score $c_{vec}^{(l)}$ is calculated as:

$$c_{vec}^{(l)} = a_{type}^{T} tanh(W_{type}\, \widehat{s}_{vec}^{(l)} + b_{type}) \tag{9}$$

where $vec \in \{R, P\}$, W_{type} and b_{type} are the weight matrix and bias vector, respectively, and a_{type}^{T} is the weight vector. The normalized attention weight $\beta_{vec}^{(l)}$ is defined as follows:

$$\beta_{vec}^{(l)} = \frac{exp(c_{vec}^{(l)})}{\sum_{j\in\{R,P\}} exp(c_j^{(l)})} \tag{10}$$

where *exp* is an exponential function. The attention-enhanced neighbor topological representation of r_i is denoted as:

$$g_{r_i}^{(l)} = \sum\nolimits_{vec\in\{R,P\}} \beta_{vec}^{(l)}\, \widehat{s}_{vec}^{(l)} \tag{11}$$

Similarly, the neighbor topological representation of protein p_j in the l-th drug-protein network $G^{(l)}$ is also learned and represented by $g_{p_j}^{(l)}$.

Learning global topological representation based on network-level attention

Given the neighbor topological representation $g_{r_i}^{(l)}$ of r_i and the neighbor topological representation $g_{p_j}^{(l)}$ of p_j, they are concatenated from beginning to end to obtain the neighbor topological representation $g^{(l)}$ of a r_i–p_j pair, $g^{(l)} = [g_{r_i}^{(l)}\ g_{p_j}^{(l)}]$.

Since interdependencies exist between the $g^{(1)}, g^{(2)}, g^{(3)}$ and $g^{(4)}$ which are separately learned from the four drug–protein networks, an attention machanism at network level is proposed to calculate the correlation γ_{lq} between $g^{(l)}$ and the q-th neighbor topological representation $g^{(q)}$,

$$\gamma_{lq} = \frac{exp(g^{(l)}(g^{(q)})^T)}{\sum_{q=1}^{4} exp(g^{(l)}(g^{(q)})^T)} \tag{12}$$

where $(g^{(q)})^T$ is the transposition vector of $g^{(q)}$. The attention-enhanced neighbor topological representation $\widehat{g}^{(l)}$ of pairwise r_i–p_j is defined as:

$$\widehat{g}^{(l)} = \sum_{q=1}^{4} \gamma_{lq} g^{(q)} + g^{(l)} \tag{13}$$

Finally, four neighbor topological representations $\widehat{g}^{(1)}, \widehat{g}^{(2)}, \widehat{g}^{(3)}, \widehat{g}^{(4)}$ are concatenated from beginning to end to obtain the global topological representation z of a r_i–p_j pair.

Pairwise multiple attribute encoding

Attribute embedding to a pair of drug and protein nodes

Based on the biological premise that if a pair of drug r_i and protein p_j interact with or are similar to more common drugs and proteins, there is a high probability that r_i interact with p_j. A strategy for obtaining pairwise r_i–p_j attribute embedding from similarities and interactions about drugs and proteins is proposed, as shown in Figure 3.

In order to obtain more comprehensive attribute embedding information of drugs and proteins, we measure the similarity of two drugs (proteins) via the interacted proteins (drugs) based on the biological premise that similar drugs are more likely to interact with similar proteins. A_i^{R-P} and A_j^{R-P} record the information of proteins that interact with r_i and r_j, respectively. The similarity R_{ij}^{sim} based on drug-protein interactions between r_i and r_j can be calculated by cosine similarity [19] as,

$$R_{ij}^{sim} = \frac{A_i^{R-P} \cdot A_j^{R-P}}{||A_i^{R-P}|| \times ||A_j^{R-P}||} \tag{14}$$

where $||\cdot||$ represents the Euclidean norm, and $\times$ denotes the multiplication operation. Similarly, the similarity P_{ij}^{sim} based on drug–protein interactions between protein p_i and p_j is calculated through the corresponding drugs that interact with p_i and p_j, respectively.

The three drug-relation matrices $R^{che}, R^{int}, R^{sim}$ form the set $U^{drug} = \{R^{che}, R^{int}, R^{sim}\}$. The protein-relation matrix set $U^{protein} = \{P^{seq}, P^{int}, P^{sim}\}$ is obtained by the three protein-relation matrices P^{seq}, P^{int} and P^{sim}. We denote the Φ_r-th relation matrix in the set U^{drug} as $U_{\Phi_r}^{drug}(1 \le \Phi_r \le 3)$ and the Φ_p-th relation matrix in $U^{protein}$ as $U_{\Phi_p}^{protein}(1 \le \Phi_p \le 3)$. The three relation matrices of drugs and the three relation matrices of proteins are combined to obtain the nine pairwise attribute embeddings. The embedded matrices are defined as:

$$Z_\Phi = \begin{bmatrix} (U_{\Phi_r}^{drug})_{i,*}||(A^{R-P})_{i,*} \\ (A^{R-P})_{j,*}^T||(U_{\Phi_p}^{protein})_{j,*} \end{bmatrix}, Z_\Phi \in \mathbb{R}^{2\times(N_r+N_p)} \tag{15}$$

$$1 \le \Phi_r \le 3; 1 \le \Phi_p \le 3; \Phi = 3 \times (\Phi_r - 1) + \Phi_p$$

where $(U_{\Phi_r}^{drug})_{i,*}$ is the i-th row of $U_{\Phi_r}^{drug}$, which represents the similarities or interactions between r_i and all drugs. The i-th row of A^{R-P} is $(A^{R-P})_{i,*}$, which records the interactions between r_i and all proteins. The interactions between p_j and all drugs are included in $(A^{R-P})_{j,*}^T$. $(U_{\Phi_p}^{protein})_{j,*}$ records the similarities or interactions between p_j and all proteins. Since context relationships exist between the nine attribute embeddings, they are stacked to obtain the final three-dimensional attribute embedding Z,

$$Z = [Z_1; Z_2; \ldots; Z_9], Z \in \mathbb{R}^{2\times(N_r+N_p)\times 9} \tag{16}$$

where ";" is a stack operation from top to bottom.

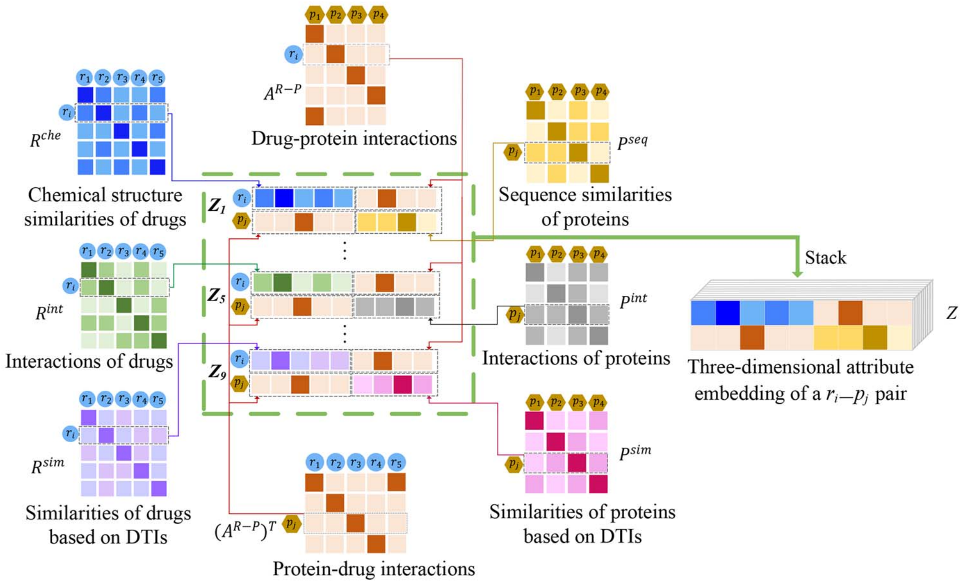

Figure 3. Illustration of the three-dimensional attribute embedding of a pair of drug and protein nodes.

Pairwise attribute encoding based on three-dimensional CNN

For the three-dimensional attribute embedding of a pair of r_i-–p_j, we construct a corresponding three-dimensional convolution neural network module to learn the relationships between features in three dimensions. The convolution module consists of two convolution-pooling layers. We take the first layer of convolution-pooling as an example to describe the process.

The attribute embedding Z is padded by zeros to preserve and learn the marginal information. The convolutional filter is denoted by $W_{conv} \in \mathbb{R}^{n_t \times n_w \times n_l \times n_h}$ where n_t, n_w, n_l and n_h are the number, width, length and depth of the filter, respectively. Z is fed into the convolutional module to encode the attribute representation of a pair of drug–protein. $Z(i,j,k)$ is the element at the i-th row, the j-th column and the k-th depth of Z. $Z_{d,i,j,k}$ is a region within the filter when the d-th filter slides to the position $Z(i,j,k)$,

$$Z_{d,i,j,k} = Z(i : i + n_w, j : j + n_l, k : k + n_h), Z_{d,i,j,k} \in \mathbb{R}^{n_w \times n_l \times n_h} \tag{17}$$

where $i \in [1, 4 - n_w + 1], j \in [1, N_r + N_p + 2 - n_l + 1], k \in [1, 11 - n_h + 1], d \in [1, n_t]$. The element value $P_{conv,d}(i,j,k)$ of the d-th feature map $P_{conv,d}$ is obtained after the d-th filter scans of $Z(i,j,k)$,

$$P_{conv,d}(i,j,k) = \delta(W_{conv}(d,:,:,:) * Z_{d,i,j,k} + b_{conv}(d)) \tag{18}$$

where $*$ is the convolution operation, b_{conv} is the bias vector, W_{conv} and δ are the weight matrix and relu function [44], repectively. We further apply a max pooling layer to extract representative features in the d-th feature map. The output is then taken as the input of the second convolution-pooling layer to obtain the final deep integrated attribute representation F of a pair of r_i-p_j.

Classification optimization and final integration

The global topological representation z of r_i-–p_j is fed as input to a fully connected layer and a softmax layer [45] to obtain the predicted interaction probability y_{topo} at topological level,

$$y_{topo} = softmax(W_{topo}z + b_{topo}) \tag{19}$$

where W_{topo} and b_{topo} are the weight matrix and bias vector, respectively. $y_{topo} = [y^0_{topo}, y^1_{topo}]$ is an interaction probability distribution of C classes (C = 2) that contains the probability y^0_{topo} indicting that r_i interacts with p_j and the probability y^1_{topo} indicting that there is no interaction between them. The loss function between the real label y_{label} and predicted interaction probability y_{topo} is cross-entropy, which is used to optimize the prediction model. The corresponding loss function $loss_1$ is defined as,

$$loss_1 = -\sum_{i=1}^{N_{train}} [y_{label} log(y^0_{topo}) + (1 - y_{label}) log(1 - y^0_{topo})] \tag{20}$$

where N_{train} denotes a set of training samples. When there is a known interaction between r_i and p_j, $y_{label} = 1$. Otherwise, $y_{label} = 0$.

The pairwise attribute representation F is flattened to obtain f, f is applied to a fully connected layer and a softmax layer to get the predicted interaction probability y_{attr},

$$y_{attr} = softmax(W_{attr}f + b_{attr}) \tag{21}$$

where W_{attr} is the weight matrix and b_{attr} is the bias vector. $y_{attr} = [y^0_{attr}, y^1_{attr}]$ where y^0_{attr} and y^1_{attr} indicate the

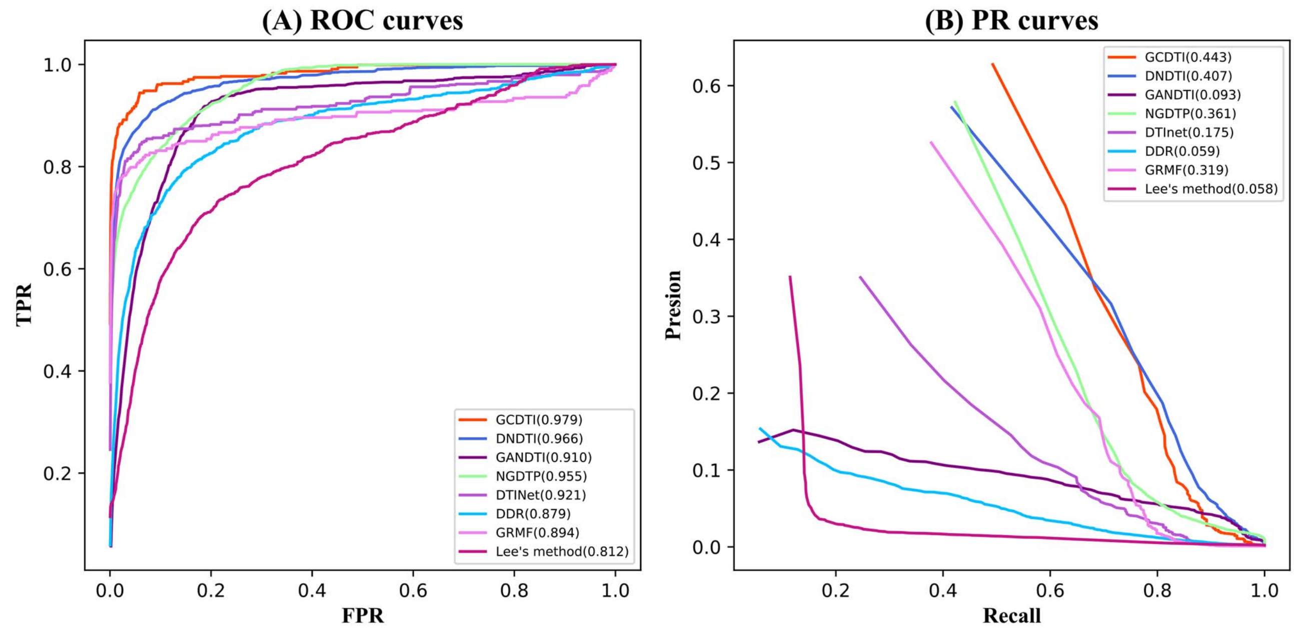

Figure 4. ROC and PR curves of all the methods for drug-protein interactions.

probability that there is an interaction or no interaction between a pair of drug-protein, respectively. The corresponding loss function is $loss_2$,

$$loss_2 = -\sum_{i=1}^{N_{train}} [y_{label} log(y^0_{attr}) + (1 - y_{label}) log(1 - y^0_{attr})] \quad (22)$$

$loss_1$ and $loss_2$ are optimized by Adam algorithm [46]. Finally, the score of interaction prediction is denoted by *score*,

$$score = \theta \times y_{topo} + (1 - \theta) \times y_{attr} \quad (23)$$

where $\theta \in [0, 1]$ is a regulation parameter that balances the contributions of global topology and pairwise attribute.

Experimental results and discussions

Evaluation metrics

Five fold cross-validation is used to evaluate the performance of the proposed method and other comparison methods. Positive samples consist of the known drug–protein interactions and the remaining unknown interactions constitute the negative samples. All positive samples are randomly divided into five equal-sized subsets, four-fifths of the positive samples are used for training, and the remaining one-fifth positive samples are used for testing. We randomly select the same number of negative samples as the positive samples for training and the remaining negative samples for testing.

When the predicted DTIs score of the sample is higher than the threshold μ, it is considered as a positive sample. Otherwise, it is regarded as a negative sample. We calculate the true positive rate (TPR) and false positive rate (FPR) as:

$$TPR = \frac{TP}{TP + FN}, FPR = \frac{FP}{TN + FP} \quad (1)$$

where TP is true positive and TN is true negative. FP is false positive and FN is false negative. Based on these metrics, ROC (Receiver Operating Characteristic) curve is drawed. The area under ROC curve (AUC) [47] is also utilized as a evaluation metric.

Due to positive and negative samples exist imbalance of distributions and the area under PR curve (AUPR) [48] contains more information than AUC, AUPR can be utilized to evaluate the prediction performance. We calculate the following metrices:

$$Precision = \frac{TP}{TP + FP}, Recall = \frac{TP}{TP + FN} \quad (2)$$

where Precision is a proportion of correct positive samples among all the predicted positive samples. Recall and TPR are the same.

Since biologists usually select the protein candidates of drugs from the top-ranked prediction results for further validation, the recall rates of top k protein candidates of drugs are calculated.

Parameter settings

The method proposed in this article is implemented using the PyTorch framework with an Nvidia GeForce GTX 2070Ti graphic card with 64 G memory. For the hyperparameters m and n in the random walk, $m = 1.3$ and $n = 0.8$. The number of performing random walk N_t is set to 10. In the global topology encoding module, the head number N_{head} of the attention mechanism of graph neural network is set to 3 and the graph neural

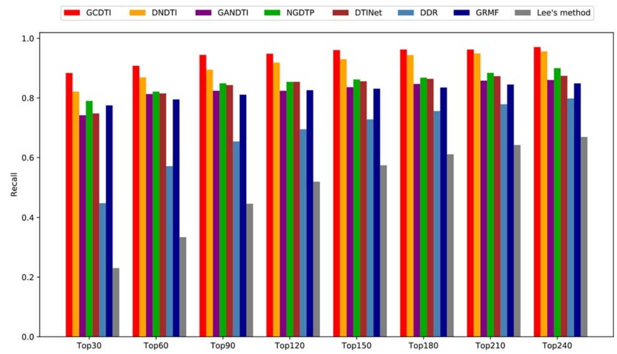

Figure 5. The recall rates of all the drugs under different top k values.

Table 1. Ablation experiment results of GCDTI

TL	NT-attention	Net-attention	AL	Average AUC	Average AUPR
✗	✗	✓	✓	0.945	0.365
✓	✗	✓	✓	0.960	0.413
✓	✓	✗	✓	0.969	0.425
✓	✓	✓	✗	0.943	0.321
✓	✓	✓	✓	**0.979**	**0.443**

network layer is set to 2. The attribute encoding module contains two convolution-pooling layers. For the first and second layer, the number of zero-padding is 1 and the number of convolution kernels is 8 and 16, respectively. The window size of the convolution kernels is 2 × 2 × 3, which respectively represent the width, length and depth of it. The value of parameter θ is set to 0.4 to integrate global topology and attribute.

Ablation experiment

We conduct a set of ablation experiments to test the independent contributions of the topological learning (TL) based on graph neural network with attention, neighbor type-level attention (NT-attention), network-level attention (Net-attention) and the attribute learning (AL) based on 3D convolutional neural network. As shown in Table 1, the final model performs best when it contains all the learning modules and the attentions. Without TL, AUC and AUPR dropped by 3.4% and 7.8% compared to the final prediction model. Without NT-attention, AUC and AUPR were 1.9% and 3.0% lower than the final model. When compared with the model without Net-attention, the final method improved the accuracy by 1.0% and 1.8% in terms of AUC and AUPR, respectively. Without AL, the performance dropped by 3.6% and 12.2% for AUC and AUPR. The experimental results demonstrate that AL has the most contribution for the improved prediction performance. It might because the AL learns the attributes of a pair of drug-protein nodes, and the attributes cover multiple kinds of similarities and interactions about drugs and proteins.

Since both drugs and proteins have multiple kinds of similarities, we conducted the ablation experiments for these similarities to evaluate their contributions for the

Table 2. Ablation experiment results for the multi-source data about the drugs and proteins

P^{seq}	P^{sim}	R^{che}	R^{sim}	P^{int}	Average AUC	Average AUPR
✗	✓	✓	✓	✓	0.947	0.381
✓	✗	✓	✓	✓	0.967	0.418
✓	✓	✗	✓	✓	0.950	0.388
✓	✓	✓	✗	✓	0.962	0.414
✓	✓	✓	✓	✗	0.953	0.391
✗	✓	✓	✓	✗	0.855	0.264
✓	✓	✓	✓	✓	**0.979**	**0.443**

prediction performance. The complete model that utilized all these similarities achieved the best performance (Table 2). The AUC and AUPR of the model without P^{seq} are 3.2% and 6.2% lower than the complete model, respectively. Without P^{sim}, AUC and AUPR decreased by 1.2% and 2.5%. Without R^{che}(or R^{sim}), AUC and AUPR dropped down by 2.9% and 5.5% (or 1.7% and 2.9%), respectively. It indicates that integrating multiple kinds of drug and protein similarities is essential for improving the prediction performance. We also performed the ablation experiment to estimate the contribution of the protein interaction data. As shown in Table 2, AUC and AUPR decreased by 2.6% and 5.2% when the protein interactions were not exploited. It demonstrates the protein interactions are important for improving the prediction performance. When both P^{seq} and P^{int} were removed, AUC and AUPR dropped down by 12.4% and 17.9%, respectively. The results indicate that utilizing both the protein sequence similarities and the protein–protein interactions is essential for improving the prediction performance. There are two primary reasons for the improved performance. First, multiple drug-protein networks contain both the protein sequence similarities and the protein interactions, which is helpful for capturing the neighbor topologies of drug and protein nodes. Second, the protein sequence similarities and the protein interactions are the important components of the attribute embedding of a protein node, which contributes to learning the attributes of the protein node.

To evaluate the contribution of each network for drug-protein interaction prediction, we performed ablation experiments on the four networks including $G^{(1)}$, $G^{(2)}$, $G^{(3)}$ and $G^{(4)}$. The experimental results are given in Table 3. When four networks were utilized, the complete model got the highest AUC (0.979) and AUPR (0.443). Compared with the model without $G^{(1)}$, the complete model's AUC and AUPR increased by 2.3% and 4.4%, respectively. Without $G^{(2)}$, AUC and AUPR decreased by 0.8% and 0.4%. Without $G^{(3)}$(or $G^{(4)}$), AUC and AUPR are 1.1% and 2.2% (or 1.4% and 3.3%) lower than the complete model. The ablation studies demonstrated that each network contributed to the improved performance of GCDTI.

Comparison with other methods

Seven advanced methods for predicting DTIs are compared with GCDTI, including DNDTI [28], GANDTI [35],

Table 3. Ablation experiment results for four networks

$G^{(1)}$	$G^{(2)}$	$G^{(3)}$	$G^{(4)}$	**Average AUC**	**Average AUPR**
✗	✓	✓	✓	0.956	0.399
✓	✗	✓	✓	0.971	0.439
✓	✓	✗	✓	0.968	0.421
✓	✓	✓	✗	0.965	0.410
✓	✓	✓	✓	**0.979**	**0.443**

NGDTP [19], DTINet [13], DDR [27], GRMF [30] and Lee's method [15]. For a fair comparison, the hyperparameters of the comparison methods are set from the values reported in the corresponding literature. (Specifically, for DNDTI, $\alpha_1 = 0.1, \alpha_2 = 0.01$. For GANDTI, $l = 500, k = 200, epoch = 2000$, learning rate $= 0.05$. For NGDTP, $\alpha_1 = 0.1, \alpha_2 = 0.1, \alpha_3 = 0.1$. For DTINet, $r = 0.8, \lambda = 1$. For DDR, $n = 600, k = 5$. For GRMF, $\eta = 0.5, d = 0.1, t = 0.1, l = 0.2$. For Lee's method, $r = 0.8$.)

As shown in Figure 4, GCDTI achieved the best average AUC of 0.979 for all comparison methods over 708 drugs, which is 1.3% higher than DNDTI and higher by 6.9%, 2.4%, 5.8%, 10.0%, 8.5% and 16.7%, for GANDTI, NGDTP, DTINet, DDR, GRMF and Lee's method, respectively. In terms of the average AUPR of all drugs, our model achieved the best AUPR of 0.443, which is 3.6%, 35.0%, 8.2%, 26.8%, 38.4%, 12.4% and 38.5% higher than those of DNDTI, GANDTI, NGDTP, DTINet, DDR, GRMF and Lee's method, respectively. To evaluate the impact of the different sizes of training dataset on GCDTI's performance, we performed 4-fold and 10-fold cross-validation. When the 10-fold cross-validation was performed, there is more the number of training samples than those in 5-fold cross-validation. As shown in Supplementary Table ST1, AUC and AUPR of ten-fold are higher 0.2% and 0.8% than five-fold. In terms of four-fold cross-validation, the number of training samples is less than 5-fold cross-validation. Compared with 5-fold cross-validation, its'AUC and AUPR dropped down by 0.1% and 0.4%. The performance of GCDTI improves with more training data. One possible reason is that more training data provides more information on drugs and proteins.

Our method GCDTI achieved the best performance in both AUC and AUPR. DNDTI and NGDTP achieved the second and third best performance among all models, taking advantage of multiple similarities between drugs and proteins. This indicates that it is important to integrate multiple similarities between drugs and proteins to improve prediction performance. The performance of DTINet and GANDTI is inferior to GRMF because they only use one similarity of drugs and one similarity of proteins. The performance of DDR and Lee's method is worse than the previous six methods. One possible reason is that DDR ignores the attribute information of drugs and proteins, and Lee's method only uses the interaction information between drugs and proteins. GCDTI achieves the best performance among all the methods, mainly because it uses multiple drug similarities and protein similarities to deeply integrate multiple attribute embeddings of drug-protein pairs and learns global topological representations from multiple heterogeneous networks.

We can obtain 708 average 5-fold AUC (AUPR) results of 708 drugs for each prediction method. 708 paired results are used to calculate Paired Wilcoxon test to compare the two methods. We present the results in Table 4. As shown, the p-value of AUCs and AUPRs is less than 0.05, which demonstrates the improvement obtained by GCDTI is statistically significant.

Figure 5 shows the recall rates of the top k protein candidates. The higher the recall rates mean more protein candidates which actually interact with drugs are correctly identified. For different cutoffs of k, the average recall rates of GCDTI are the highest with 88.3% for top 30, 90.8% for top 60 and 94.4% for top 90. The recall rates of DNDTI are inferior to our model. When k is 30, 60 and 90, they are 82.1%, 86.9% and 89.5% respectively. The third best performance method is NGDTP with recall rates of 79.0%, 82.1% and 84.9% for top 30, 60 and 90 respectively. The recall rates of DTINet are close to GANDTI where the recall rates of former are 74.8%, 81.5%, 84.3%, and the recall rates of latter are 74.2%, 81.3% and 82.4%, for the top 30, 60 and 90 protein candidates, respectively. For GRMF, the recall rates are 77.5%, 79.5% and 81.1%. DDR has lower recall rates of 44.8%, 57.1% and 65.4%, and Lee's method has the worst recall rates of 23.0%, 33.4%, 44.6%, for top 30, 60 and 90, respectively.

We listed the average training and testing time of all the methods in supplementary table ST2. GCDTI's average training time for each fold of cross-validation is 15.2 minutes. The slowest model is NGDTP which takes 20.4 minutes, and another model (DNDTI) takes 16.8 minutes. When each drug–protein node pair is tested, our method takes average 98.2 microseconds for it. NGDTP, DDR and Lee's method takes 1001.8 ms, 195.3 ms and 25.3 ms, respectively. The results demonstrate that the average training and testing efficiencies of these methods are not significantly different.

Case studies on five drugs

Through case studies of Olanzapine, Ziprasidone, Amoxapine, Asenapine and Paroxetine, we can further verify GCDTI's ability to discover potential DTIs. We sort all protein candidates for each drug in descending order according to the predicted interaction scores. The top ten protein candidates for each drug are listed in Table 5.

Drugbank is a database of molecular information which contains detailed drug data, target information and the corresponding mechanisms of interactions [38]. DrugCentral provides information on the interactions between drugs and proteins, mechanisms of drug action, indications and pharmacological effects [49]. For a total of 50 candidate proteins, Drugbank and DrugCentral recorded 32 and 37 candidate proteins, respectively. This demonstrates that these candidate proteins have an interaction with the corresponding

Table 4. The statistical results of the paired wilcoxon test on AUCs and AUPRs between GCDTI and other methods

	DNDTI	GANDTI	NGDTP	DTINet	DDR	GRMF	Lee's method
p-value of AUC	1.28005e-15	1.88951e-08	8.84386e-43	3.70013e-04	3.11427e-06	4.80001e-03	5.17616e-09
p-value of AUPR	8.82913e-14	5.25370e-05	3.13560e-08	3.56269e-20	2.10539e-12	4.75020e-03	1.05591e-07

Table 5. The top 10 target candidates of 5 drugs

Drug name	Rank	Target	Evidence	Rank	Target	Evidence
	1	HTR1A	DrugBank/DrugCentral	6	ADRA1A	DrugBank/DrugCentral
	2	HTR2B	DrugCentral	7	ADRB1	DrugBank/DrugCentral
Olanzapine	3	CHRM1	DrugBank/DrugCentral	8	HTR1F	DrugBank/DrugCentral
	4	HTR1D	DrugBank/DrugCentral	9	ADRB2	DrugBank/DrugCentral
	5	ADRA1D	DrugCentral	10	GABRA1	DrugBank
	1	KCNH2	DrugCentral	6	SLC6A4	DrugCentral
	2	HTR1D	DrugBank/DrugCentral	7	ADRB1	DrugCentral
Ziprasidone	3	HTR2B	DrugCentral	8	HRH2	DrugCentral
	4	CHRM1	DrugBank/DrugCentral	9	ADRB2	DrugCentral
	5	ADRA2C	DrugBank/DrugCentral	10	SLC6A2	DrugCentral
	1	ADRA2B	DrugBank/DrugCentral	6	ADRA1B	DrugBank
	2	DRD3	DrugBank/DrugCentral	7	CHRM5	DrugBank/DrugCentral
Amoxapine	3	CHRM4	DrugBank/DrugCentral	8	ADRA1D	DrugBank/DrugCentral
	4	ADRA1A	DrugBank	9	HTR7	DrugBank/DrugCentral
	5	HTR2A	DrugBank/DrugCentral	10	CHRM3	DrugBank/DrugCentral
	1	CHRM3	DrugCentral	6	ADRA1D	Literature [51]
	2	ADRA1B	DrugCentral	7	ADRA2A	DrugBank/DrugCentral
Asenapine	3	ADRA2B	DrugBank/DrugCentral	8	CHRM5	DrugCentral
	4	SLC6A4	Literature [50]	9	CHRM1	DrugCentral
	5	HTR3A	DrugCentral	10	DRD5	DrugCentral
	1	HTR2C	DrugBank	6	ADRA2C	DrugBank
	2	HTR1A	DrugBank/DrugCentral	7	DRD2	DrugBank
Paroxetine	3	ADRA1A	DrugBank	8	HTR7	DrugBank
	4	HTR6	DrugBank	9	SLC6A3	DrugCentral
	5	ADRA2A	DrugBank	10	ADRA2B	DrugBank

drugs. Two candidate proteins labelled as "literature" are supported by published papers [50, 51], indicating the strong possibility of the proteins interacting with these drugs. All case studies show that GCDTI can indeed detect potential drug-related protein candidates.

Prediction of novel drug–protein interactions

We apply GCDTI to predict the candidate proteins that interact with each drug. Known drug–protein interactions are utilized to train GCDTI prediction model. The predicted top 10 candidate proteins of each drug are recorded in the Supplementary Table ST3 to help biologists discover potential DTIs in wet-lab experiments.

Conclusion

We propose a method to integrate and encode similarities and interactions within multiple heterogeneous networks to predict candidate drug-related proteins. The proposed pairwise embedding strategy can integrate the inter- and intra relations between drugs and proteins. The established multiple drug–protein networks benefit the formation of multi-type neighbor node sequences based on random walk. We construct a framework based on graph neural network with attention and multilayer three-dimensional convolutional neural network to encode and learn global topological representations and attribute representations. Three attention mechanisms are proposed to assign higher weights to more important neighbor nodes, neighbor types and neighbor topologies. The comparison with seven drug–protein interaction prediction methods shows that the proposed method improves the performance on AUC and AUPR. GCDTI is attractive to biologists as more true drug–protein interactions are usually contained in its top-k candidates. Case studies of five drugs further confirm the powerful capability of the proposed method to discover potential DTIs candidates.

Key Points

- Four double-layer heterogeneous networks are established to benefit the integration and extraction of similarities and interactions related to drugs and proteins from multiple data sources.
- A novel framework based on graph neural network and three-dimensional convolutional neural network encodes two levels of representations of drug–protein

pairs, where the global topological representation reveals neighbor topological structure of drugs and proteins in multiple heterogeneous networks, whereas the attribute representation reveals underlying and deep relations between drugs and proteins.

- New attention mechanisms at neighbor node level, neighbor type level and network level discriminate the different contributions of neighbor node, neighbor type and neighbor topology for the final drug–protein interaction prediction, respectively.
- The improved prediction performance is verified by the comparison with seven advanced methods over public datasets, recall rates of top *k* candidates, case studies of five drugs.

Supplementary information

Supplementary data are available at *Briefings in Bioinformatics* online.

Data availability

The dataset about this study is available at https://github.com/pingxuan-hlju/GCDTI.

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References

1. Chen X, Yan CC, Zhang X, *et al.* Drug-target interaction prediction: databases, web servers and computational models. *Brief Bioinform* 2015;**17**(4):696–712.
2. Zheng S, Li Y, Chen S, *et al.* Predicting drug-protein interaction using quasi-visual question answering system. *Nat Mach Intell* 2020;**2**(2):134–40.
3. Wu L, Shen Y, Li M, *et al.* Network output controllability-based method for drug target identification. *IEEE Trans Nanobiosci* 2015;**14**(2):184–91.
4. Mathur A, Loskill P, Shao K, *et al.* Human iPSC-based cardiac microphysiological system for drug screening applications. *Sci Rep* 2015;**5**(1):1–7.
5. Hu SS, Zhang C, Chen P, *et al.* Predicting drug-target interactions from drug structure and protein sequence using novel convolutional neural networks. *BMC Bioinform* 2019;**20**(25):1–12.
6. Wang L, You ZH, Chen X, *et al.* Computational methods for the prediction of drug-target interactions from drug fingerprints and protein sequences by stacked auto-encoder deep neural network. *Int Symp Bioinform Res Appl* 2017;**10330**:46–58.
7. Ding Y, Tang J, Guo F. Identification of drug-target interactions via multiple information integration. *Inform Sci* 2017;**418-419**:546–60.
8. Keiser MJ, Roth BL, Armbruster BN, *et al.* Relating protein pharmacology by ligand chemistry. *Nat Biotechnol* 2007;**25**(2):197–206.
9. Keiser MJ, Setola V, Irwin JJ, *et al.* Predicting new molecular targets for known drugs. *Nature* 2009;**462**(7270):175–81.
10. Morris GM, Huey R, Lindstrom W, *et al.* AutoDock4 and AutoDockTools4: automated docking with selective receptor flexibility. *J Comput Chem* 2009;**30**(16):2785–91.
11. Li Y, Huang Y-A, You Z-H, *et al.* Drug-target interaction prediction based on drug fingerprint information and protein sequence. *Molecules* 2019;**24**(16):2999.
12. He S, Guo F, Zou Q. MRMD2. 0: a python tool for machine learning with feature ranking and reduction. *Curr Bioinform* 2020;**15**(10):1213–21.
13. Luo Y, Zhao X, Zhou J, *et al.* A network integration approach for drug-target interaction prediction and computational drug repositioning from heterogeneous information. *Nat Commun* 2017;**8**(1):573.
14. Chen X, Liu M-X, Yan G-Y. Drug-target interaction prediction by random walk on the heterogeneous network. *Mol Biosyst* 2012;**8**(7):1970–8.
15. Lee I, Nam H. Identification of drug-target interaction by a random walk with restart method on an interactome network. *BMC Bioinform* 2018;**19**(8):208.
16. Bleakley K, Yamanishi Y. Supervised prediction of drug-target interactions using bipartite local models. *Bioinformatics* 2009;**25**(18):2397–403.
17. Keum J, Nam H. SELF-BLM: Prediction of drug-target interactions via self-training SVM. *PLoS One* 2017;**12**(2):0171839.
18. Xuan P, Sun C, Zhang T, *et al.* Gradient boosting decision tree-based method for predicting interactions between target genes and drugs. *Front Genet* 2019;**10**:459.
19. Xuan P, Chen B, Zhang T, *et al.* Prediction of drug-target interactions based on network representation learning and ensemble learning. *IEEE/ACM Trans Comput Biol Bioinform* 2021;**18**(6):2671–81.
20. Ru X, Ye X, Sakurai T, *et al.* Current status and future prospects of drug-target interaction prediction. *Brief Funct Genomics* 2021;**20**(5):312–22.
21. Niu M, Zou Q, Wang C. GMNN2CD: Identification of circRNA-disease associations based on variational inference and graph markov neural networks. *Bioinformatics* 2022;**2**:1–10.
22. Zou Y, Ding Y, Peng L, *et al.* FTWSVM-SR: DNA-binding proteins identification via fuzzy twin support vector machines on self-representation. In: Wei D (ed). *Interdiscip Sci Comput Life Sci*, Germany, 2021, 1–13.
23. Wen M, Zhang Z, Niu S, *et al.* Deep-learning-based drug-target interaction prediction. *J Proteome Res* 2017;**16**(4):1401–9.
24. Xuan P, Zhang Y, Cui H, *et al.* Integrating multi-scale neighbouring topologies and cross-modal similarities for drug-protein interaction prediction. *Brief Bioinform* 2021;**22**(5):1–8.
25. Lee I, Keum J, Nam H. DeepConv-DTI: Prediction of drug-target interactions via deep learning with convolution on protein sequences. *PLoS Comput Biol* 2019;**15**(6):1–21.
26. Rifaioglu AS, Nalbat E, Atalay V, *et al.* DEEPScreen: high performance drug-target interaction prediction with convolutional neural networks using 2-D structural compound representations. *Chem Sci* 2020;**11**(9):2531–57.
27. Olayan RS, Ashoor H, Bajic VB. DDR: efficient computational method to predict drug-target interactions using graph mining and machine learning approaches. *Bioinformatics* 2017;**34**(7):1164–73.

28. Xuan P, Hu K, Cui H, *et al*. Learning multi-scale heterogeneous representations and global topology for drug-target interaction prediction. *IEEE J Biomed Health Inform* 2021;**10**:1–12.
29. Zheng X, He S, Song X, *et al*. DTI-RCNN: new efficient hybrid neural network model to predict drug-target interactions. *Artif Neural Networks Mach Learn ICANN* 2018;**2018**:104–14.
30. Ezzat A, Zhao P, Wu M, *et al*. Drug-Target Interaction Prediction with Graph Regularized Matrix Factorization. *IEEE/ACM Trans Comput Biol Bioinform* 2016;**14**:1–1.
31. Zhao T, Hu Y, Valsdottir LR, *et al*. Identifying drug-target interactions based on graph convolutional network and deep neural network. *Brief Bioinform* 2021;**22**(2):2141–50.
32. Wang YB, You ZH, Yang S, *et al*. A deep learning-based method for drug-target interaction prediction based on long short-term memory neural network. *BMC Med Inform Decis Mak* 2020;**20**(2):1–9.
33. Chen ZH, You ZH, Guo ZH, *et al*. Prediction of drug-target interactions from multi-molecular network based on deep walk embedding model. *Front Bioeng Biotechnol* 2020;**8**:338.
34. Wang W, Lv H, Zhao Y, *et al*. DLS: a link prediction method based on network local structure for predicting drug-protein interactions. *Front Bioeng Biotechnol* 2020;**8**:330.
35. Sun C, Xuan P, Zhang T, *et al*. Graph convolutional autoencoder and generative adversarial network-based method for predicting drug-target interactions. *IEEE/ACM Trans Comput Biol Bioinform* 2022;**19**(1):455–64.
36. Manoochehri HE, Nourani M. Drug-target interaction prediction using semi-bipartite graph model and deep learning. *BMC Bioinform* 2020;**21**(4):1–16.
37. Zhao Y, Zheng K, Guan B, *et al*. DLDTI: a learning-base d framework for drug-target interaction identification using neural networks and network representation. *J Transl Med* 2020;**18**(1):1–15.
38. Wishart D, Djoumbou Y, Guo AC, *et al*. DrugBank 5.0: a major update to the DrugBank database for 2018. *Nucleic Acids Res* 2018;**46**:D1074–82.
39. Keshava Prasad TS, Goel R, Kandasamy K, *et al*. Human protein reference database–2009 update. *Nucleic Acids Res* 2009;**37**(Database issue):D767–72.
40. Iorio F, Bosotti R, Scacheri E, *et al*. Discovery of drug mode of action and drug repositioning from transcriptional responses. *Proc Natl Acad Sci* 2010;**107**(33):14621.
41. Wang W, Yang S, Zhang X, *et al*. Drug repositioning by integrating target information through a heterogeneous network model. *Bioinformatics* 2014;**30**(20):2923–30.
42. Grover A, Leskovec J. node2vec: Scalable feature learning for networks. *KDD* 2016;**2016**:855–64.
43. Maas AL, Hannun AY, Ng AY. Rectifier nonlinearities improve neural network acoustic models. *Proc icml* 2013;**30**(1):3.
44. Nair V, Hinton GE. Rectified linear units improve restricted boltzmann machines. In: Fürnkranz J, Joachims T (eds). *Proceedings of the 27th International Conference on International Conference on Machine Learning*. International Conference on Machine Learning (ICML), Haifa, Israel, 2010, 807–14.
45. Bahdanau D, Cho K, Bengio Y. Neural machine translation by jointly learning to align and translate. In: Bengio Y, LeCun Y (eds). *International Conference on Learning Representations*. International Conference on Learning Representations (ICLR), San Diego, CA, USA, 2015, abs/1409.0473.
46. Kingma D, Ba J. Bengio Y, LeCun Y (eds). *Adam: A Method for Stochastic Optimization. International Conference on Learning Representations*. International Conference on Learning Representations (ICLR), San Diego, CA, USA, 2014.
47. Hajian-Tilaki K. Receiver operating characteristic (ROC) curve analysis for medical diagnostic test evaluation. *Caspian J Intern Med* 2013;**4**(2):627–35.
48. Saito T, Rehmsmeier M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PloS One* 2015;**10**(3):1–21.
49. Ursu O, Holmes J, Knockel J, *et al*. DrugCentral: online drug compendium. *Nucleic Acids Res* 2017;**45**(D1):D932–9.
50. Bosc N, Atkinson F, Felix E, *et al*. Large scale comparison of QSAR and conformal prediction methods and their applications in drug discovery. *J Chem* 2019;**11**(1):4.
51. McIntyre RS. Asenapine: a review of acute and extension phase data in bipolar disorder. *CNS Neurosci Ther* 2011;**17**(6):645–8.